# Data Scraping & Trust Scoring

Assignment Report

*Multi-Source Scraper + Content Trust Scoring System*

## 1. Project Overview

This project builds an end-to-end pipeline that scrapes content from three source types- blogs, YouTube, and PubMed, then evaluates each source using a five-factor trust scoring algorithm. Every source produces a structured JSON object with metadata, content chunks, topic tags, language, and a trust score with a full factor-by-factor breakdown.

- **Sources:** 3 blogs | 2 YouTube videos | 1 PubMed article
- **Output:** output/final_output.json - one object per source
- **Trust range:** 0.0 (no extractable data) → 1.0 (maximum credibility)

### Pipeline Overview

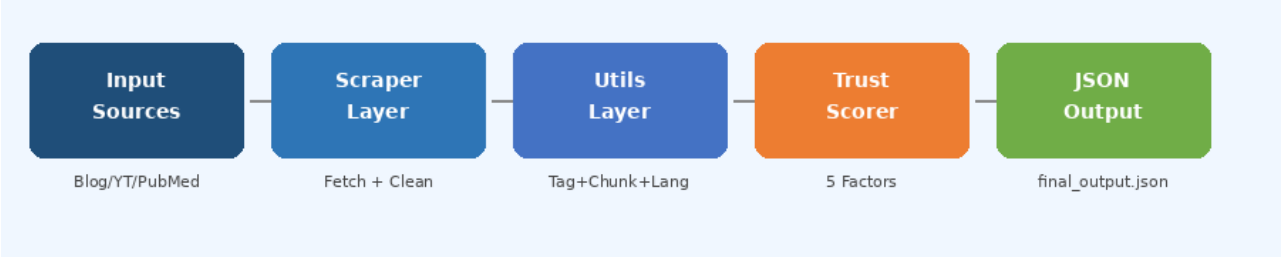


*Fig 1 — Five-stage pipeline from input sources to JSON output*

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## 2. Scraping Strategy

### 2.1 Blog Scraper (requests + BeautifulSoup4)

- **Fetch page** - requests.get() with browser User-Agent to avoid 403 blocks
- **Remove noise** - strips nav, header, footer, aside, ads, sidebar elements
- **Find article body** - looks for <article> → <main> → body (priority order)
- **Extract content** - paragraphs longer than 40 chars only
- **Extract author** - meta tags → JSON-LD structured data → HTML byline
- **Extract date** - article:published_time → JSON-LD → <time> tag
- **Clean author** - strips @ prefixes, URLs, rejects generic names like 'Web Author'

### 2.2 YouTube Scraper (youtube-transcript-api v0.6+)

- Instantiates YouTubeTranscriptApi() and calls .fetch(video_id)
- Joins all transcript segments into a single content string

- On failure → content = 'Transcript not available', language = 'unknown'
- Trust scorer applies content penalty for missing transcripts
- No API key required - pulls from YouTube's public endpoints

### 2.3 PubMed Scraper (Entrez XML API + Semantic Scholar)

- Calls eutils.ncbi.nlm.nih.gov/efetch with the PMID, parses XML response
- Extracts: title, abstract, all authors, journal name, publication date
- Multiple authors joined as comma-separated string for trust scorer averaging
- Date handles both electronic (ArticleDate) and print (PubDate) formats
- Month abbreviations converted: Jan → 01, Feb → 02, etc.
- Citation count: tries PubMed XML → falls back to Semantic Scholar free API

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## 3. Topic Tagging Method (spaCy en_core_web_sm)

| Step | Action | Output |
|---|---|---|
| 1 | Run content through spaCy NLP pipeline | Tokenised, POS-tagged document |
| 2 | Extract all NOUN tokens (lowercased) | List of common nouns from text |
| 3 | Extract all named entities (lowercased) | Proper nouns, organisations, places |
| 4 | Merge both lists, count frequency | word → count dictionary |
| 5 | Return top 5 most frequent terms | topic_tags array in output JSON |

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## 4. Trust Score Algorithm

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Trust Score = author_credibility + citation_count + domain_authority +
              recency + medical_disclaimer + source_and_content
```

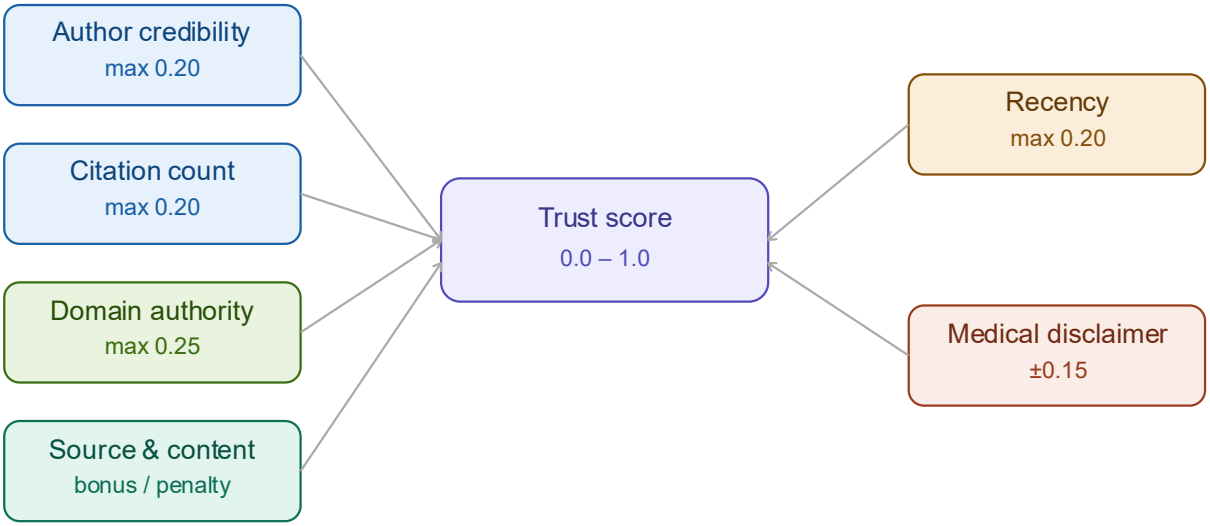


*Fig 2 — Five factors feeding into the final trust score (0.0 – 1.0)*

| Factor | Max | Scoring logic | Scores 0 when |
|---|---|---|---|
| Author Credibility | 0.20 | +0.10 named person present. +0.05 if PhD/MD/Dr in name. Averaged across multiple authors. Penalised for short single-word names | Name is "Web Author", "YouTube Channel", "admin", etc. |
| Citation Count | 0.20 | +0.20 ≥100 citations. +0.15 ≥50. +0.10 ≥10. +0.05 ≥1. For blogs: infers from DOI count and reference section mentions | No citations in XML and Semantic Scholar returns nothing |
| Domain Authority | 0.25 | Static lookup table (pubmed=1.0, ibm=0.85, geeksforgeeks=0.60). Bonus for .edu (+0.18) and .gov (+0.20) domains | Spam keywords in URL path, or completely unknown domain |
| Recency | 0.20 | +0.20 if ≤1yr old. +0.15 if ≤2yr. +0.10 if ≤3yr. +0.05 if ≤5yr. Medical content >3yr gets extra -0.10 penalty | Date is missing ("Unknown") or article is older than 5 years |
| Medical Disclaimer | +0.15 / -0.10 | Only triggers if ≥3 medical keywords found. +0.15 if disclaimer phrase present. -0.10 if no disclaimer. Returns 0 for non-medical content | Content is not medical (neutral — no bonus, no penalty) |
| Source & Content | +0.20 / penalty | +0.15 if source_type=pubmed. +0.08 if youtube or blog. +0.05 if content >3000 chars, +0.03 if >1000 chars. −0.05 if <100 chars. +0.02 if language is known. −0.05 if YouTube transcript unavailable | Short content (<100 chars) with no language set and missing YouTube transcript |

## 5. Edge Case Handling

| Scenario | Handling |
|---|---|
| Missing author | author_credibility = 0. No crash - pipeline continues |
| Multiple authors (PubMed) | All extracted, joined as string, credibility scores averaged |
| Missing publish date | recency = 0. Date field = "Unknown" in output |
| Non-English content | Language detected and stored. Scoring works regardless of language |
| Content < 100 characters | Entry skipped entirely — too short to be useful |
| Twitter handle as author (@Simplilearn) | _clean_author() strips @ prefix before storing name |
| JavaScript-rendered page | Returns < 100 chars → skipped with console warning |
| YouTube transcript unavailable | Content = "Transcript not available". language = "unknown". Content penalty applied |
| PubMed citation count = 0 in XML | Semantic Scholar API called as fallback. Returns "0" if both fail |
| Medical content, no disclaimer | -0.10 penalty applied to flag potentially misleading health info |

## 6. Abuse Prevention Logic

| Abuse Type | Detection | Penalty |
|---|---|---|
| Fake / generic authors | Name matches known placeholder list: "Web Author", "admin", "editor", "staff", single-word names <4 chars | author_credibility = 0 |
| SEO spam blogs | Spam keywords detected in URL path: "free-traffic", "seo-hack", "click-here", "buy-now" | domain_authority forced to 0 regardless of domain |
| Misleading medical content | ≥3 medical keywords in content AND no recognised disclaimer phrase found | -0.10 medical_disclaimer penalty |
| Outdated information | Article older than 5 years → recency = 0. Medical content older than 3 years gets additional -0.10 | Combined recency and medical penalty can significantly lower score |

## 7. Results & Score Analysis

| Source | Type | Score | Strong factors | Limiting factors |
|---|---|---|---|---|
| IBM - Machine Learning | Blog | 0.56 | High domain authority (0.21), strong recency (0.20), valid author | No citations, medical keyword penalty (-0.10) |
| GeeksforGeeks - ML | Blog | 0.58 | Good recency (0.15), decent content quality | Moderate domain authority (0.15), no citations |
| IBM - Neural Networks | Blog | 0.66 | High domain authority (0.21), strong recency (0.20), clear content | No citation support |
| YouTube #1 | YouTube | 0.25 | Basic domain authority (0.10), some content extracted | Missing metadata (author, date), no citations |
| YouTube #2 | YouTube | 0.25 | Basic domain authority (0.10), partial transcript | Missing metadata (author, date), no citations |
| PubMed Article | PubMed | 0.65 | Highest domain authority (0.25), multiple authors, structured research content | Low citation count, slightly older publication |

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## 8. Limitations & Future Improvements

| Limitation | Root cause | Future fix |
|---|---|---|
| JS-rendered sites skipped | requests cannot execute JavaScript | Add Playwright/Selenium for dynamic rendering |
| Static domain authority table | Live APIs (Moz, Ahrefs) cost $50–200/month | Integrate free OpenPageRank API |
| Citation count unreliable | PubMed XML field not always populated | Already using Semantic Scholar as fallback |
| Frequency-based topic tags | Noun frequency is simple but imprecise | Switch to KeyBERT for semantic keyword extraction |
| YouTube transcripts blocked | Network-level restriction in test environment | VPN/proxy or YouTube Data API v3 with key |